The Nexus Fold Compiler

Binary Ring Algebra, Register Allocation, and Co-Translational Address Creation toward De Novo Protein Folding

Dean A. Kulik, QuHarmonics Research Group

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Ontological Architecture: The Polypeptide as an Executable Route Through Native Binary Rings

Conventional structural bioinformatics treats a protein as a static chemical noun—a linear polymer of amino acids that subsequently searches a rugged thermodynamic energy landscape to locate a three-dimensional minimum. Modern deep-learning predictors such as AlphaFold bypass physical simulation by learning an empirical mapping from millions of homologous sequence alignments to spatial coordinates. While statistically powerful, empirical pattern matching treats the folding process as a black-box lookup table rather than a deterministic computation.

The NEXUS fold compiler establishes the exact opposite paradigm: transformations constitute what physically exists, verbs precede nouns, and a protein is not an object that later gets folded, but a deterministic execution route through a pre-existing relational memory map. In this transformation-first ontology, denoted by the foundational identity Co ≡ C1, a primitive constraint generates a binary distinction and a constrained continuation; propagation across that continuation retains history, establishing a fixed relational structure whose physical geometry is a reader-dependent observation. The three-dimensional fold is the spatial readout—the physical residual shape or structural cavity—left behind by executing that route. Rather than inferring the map from evolutionary examples, the compiler takes the relational memory map as given by native binary rings and seeks the exact algebraic reader capable of extracting spatial addresses directly from the linear sequence input, analogous to how the Bailey–Borwein–Plouffe (BBP) formula extracts the n-th hexadecimal digit of π without computing preceding digits, or how the SHA-256 compression function routes state transitions through fixed Boolean registers.

Central to this systems architecture is the strict separation between the program and its execution hardware:

The code that runs ≠ The code that runs the code

The genetic and polypeptide sequence is the *code that runs*—an incomplete symbolic expression analogous to the partial instruction 2 + =, which fixes the operator and the first operand but cannot evaluate until a second operand arrives. The surrounding physical field—comprising water molecules, pH, thermal scheduling, the ribosomal stream loader, lipid membranes, and chaperone exception handlers—is the *code*

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that runs the code (the runtime execution environment and relational CPU) that dynamically supplies the second operand at each step of chain emission.

Every architectural hypothesis within the NEXUS compiler is audited as executable code against experimental protein structures without fitted energy weights, governed by a strict four-state verification ledger and cryptographic pre-registration via SHA-256 hashes generated prior to opening validation datasets:

Ledger State	Compiler Verification Definition	Operational Consequence
RUNS	The algorithmic route computed the target spatial or algebraic address prior to linking empirical data, and the experimental data landed on the prediction.	Promoted to core compiler pipeline; routes sealed before their landing data carry a SHA-256 hash.
STANDS	The address, symmetry, or thermodynamic boundary was supplied by an established external theorem or measured physical constant.	Retained as verified structural ground and boundary constraint.
DEAD END	The code executed and produced a clean landing, but the output belongs to a different architectural layer or owner rather than the target variable.	Falsified for the immediate target; routed back to its true structural owner to open the next investigative lane.
OPEN	The exact topological shape of the missing link is bounded by reverse-the-gap subtraction, but the sequence-side reader or 3D bridge is not yet written.	Isolated as an explicit engineering specification for subsequent compiler builds.

Decompilation Limits, Self-Clocking Streams, and Hairpin Network Address Translation

The 6-Bit Decompilation Baseline and the Local Window Wall

At the geometric level, a linear sequence of 20 canonical amino acid types compiles into a closed three-dimensional backbone topology whose stability is dominated by nonlocal contacts—residues distant in

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sequence index (i, j) converging in physical space. In early compiler builds, real protein backbones were decompiled into a discrete 6-bit instruction per residue step—comprising 2 alpha-carbon (Cα) bend bits and 4 torsion twist bits, mapping directly onto the 64 six-bit words of the native ring R_6—and recompiled into 3D coordinates by taking the running product of spatial rotation matrices. While a 6-bit instruction word per step reconstructs a 20-residue polypeptide chain faithfully, longer chains accumulate rotational drift and require 12 to 14 bits per step to hold the same reconstruction accuracy. By contrast, the upstream genetic layer executes with zero loss: reading bacterial genes through a 2-bit-per-base codon reader yields their corresponding amino acid sequences with zero mismatches.

When readers predict local secondary structure states (alpha-helix, beta-strand, or coil) by scanning a short sliding window of residues, they stall near the same place: the compiler's own window parsers classify 51.5% and 51.6% of residues correctly. Readers that take the whole sequence at once—still from a single sequence, with no alignment—reach 72.5% (SPIDER3-Single; Heffernan et al., 2018) and about 74% (SPOT-1D-Single; Singh et al., 2021). What separates the two is whole-chain context. Treating the system as code shows why the window stalls: a major component of the three-dimensional fold—specifically the nonlocal return route across antiparallel structures—is not encoded within the static values of a local window at all, but is dynamically created as an execution artifact during compilation.

Self-Clocking Streams and the Palindrome Pinch on R_n

To understand how a linear molecular stream encodes structural turns without an external clock, the compiler draws directly from asynchronous signal decoding, specifically the SonicDMX audio-lighting decoder architecture. In an unclocked binary stream such as 110011, a receiver cannot distinguish how many consecutive identical bits (11 or 00) have arrived because static repeats generate no state transitions. By introducing a tertiary repeat symbol 2 defined as *"the same value as the immediately preceding symbol,"* the sequence 110011 is recoded into 120212. Under this transformation, every emitted symbol is guaranteed to differ from its predecessor, making every step a state transition that carries its own synchronization tick. Because no symbol ever repeats consecutively, each transition chooses between only two remaining states on a three-vertex triangle, encoding all stream information purely in the rotational direction of travel.

The native binary ring $R_n = \{0,1\}^n$ (of cardinality $N = 2^n$) executes this exact self-clocking logic through its facing law:

$$R[N - 1 - a] = P_n(R[a])$$

For almost all words on the ring, the native partner operator P_n acts by bit-reversal J, mapping $b_1 b_2 \ldots b_n \mapsto b_n \ldots b_2 b_1$. Because reversal is bijective and deterministic on asymmetric words, once one end of a reversal pair is emitted, its partner is fixed without ambiguity. However, if a word is a **palindrome** ($w = J(w)$), bit-reversal leaves the word unchanged, producing a static duplicate that would stall relational continuation. Precisely at palindromic fixed points, the native ring automatically switches its partner operator from reversal J to bitwise complement C ($w \mapsto \bar{w}$). The ring injects its own "symbol 2" exactly where a self-duplicate would occur.

Topologically, because a palindrome cancels under reversal, it represents a zero-width **pinch** where the outgoing forward route and the returning backward route meet. In a polypeptide chain, this palindromic pinch manifests physically as a beta-hairpin turn (the crease), while the paired antiparallel beta-strands

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extending away from the turn form the pinched seam, analogous to folding a dumpling at a central crease and pinching its opposing edges together.

Hairpin NAT and the Packet-vs.-Table Falsification

Translated into network systems engineering, a protein beta-hairpin functions as **Hairpin NAT (Network Address Translation)**. A data packet exits a local network on a public sequence address (residue index i) and must loop back internally to a private spatial destination address (contacting residue index j). For the return packet to be accepted rather than dropped, the NAT router must rewrite both the destination address and the return path simultaneously. In a protein hairpin, a single integer invariant executes both rewrites at once—the crease sum S:

$$S = i + j \Longrightarrow j = S - i$$

Fixing S simultaneously determines the partner address $j = S - i$ and inverts the traversal direction ($\Delta j = -\Delta i$).

Experimental Look-Back & NAT Run	**Dataset & Configuration**	**Quantitative Output**	**Ledger Status & Architectural Owner**
Free-Edge Strand Pairing	30 training structures using true secondary structure labels; pair each new strand with the most recent strand holding an open edge.	27 real strand pairs found, 49 wrong calls, 25 missed (precision 0.36, recall 0.52; legacy nearest-within-8 rule: 20 found, 27 wrong, precision 0.43); strand direction predicted from the intervening loop in 24 of 27 found pairs (88.9%).	**RUNS** for direction; **OPEN** for partner selection when skipping strands.
Whole-Chain Facing Check	30 training structures; test whether residue i contacts whole-chain partner $N - 1 - i$ globally.	428 contacts on global facing line vs. 516 expected by chance; mutual information = 0.010 bits vs.	**DEAD END**; N-to-C terminal pairing in GB1 and Protein L is a fold-family trait, not a

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Experimental Look-Back & NAT Run	Dataset & Configuration	Quantitative Output	Ledger Status & Architectural Owner
		0.011 bits random.	universal memory law.
Deferred Encounter Decoder	Sealed run (NEXUS_deferred_decoder_run.zip) on 4 exact genes: GB1, Protein L, thioredoxin (K02845), barstar (X15545).	Holding ambiguous segments pending until hairpin encounter improved state labels on only 2 of 4 proteins (bar $\geq 3/4$).	**DEAD END**; local ambiguity resolution requires active field open-slot pressure.
Hydropathy Flip Stream	Binary water-loving vs. water-avoiding transition stream across training segments.	Strands flip at 0.52 rate vs. helices at 0.48; strands sustain longer steady runs (0.42 vs. 0.28 of segment length).	**OPEN** (weak statistical signal; insufficient to call creases alone).
Codon Middle-Base Mirror	108 real beta-hairpins (leave-one-out cross-validation) testing mirror symmetry of codon middle base across the crease.	Like pairs with like in folded sheets, but mirror term adds zero crease detection power ($7/101$ vs. $8/100$ without mirror term).	**DEAD END** for sequence values; the return address is not stored in the packet.

The failure of the codon middle-base mirror test (7/101 vs. 8/100) and the whole-chain facing check exposes a fundamental systems principle established by the Hairpin NAT model. In network routing, a NAT router never finds the return address written inside the payload of the outgoing packet; the router writes the return entry into its own internal state translation table at the moment the packet leaves. Any bioinformatics reader that scans linear sequence values looking for a pre-written mirror of the returning strand is inspecting the packet payload for routing information that only the runtime state table holds.

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The Genetic Code as an Executable Lexer on the R_6 Hypercube

Binary Base Geometry and the 6-Dimensional Boolean Cube

Rather than treating the genetic code as an arbitrary frozen biological lookup table, decompiling the nucleotide bases into their underlying chemical switches reveals an exact 6-bit Boolean hypercube (R_6, with $2^6 = 64$ vertices). Each nucleotide base is completely specified by a 2-bit shape word $b_{\text{high}}b_{\text{low}}$. Specifically, the high bit (size or ring count) assigns 0 to purines (two rings: Adenine, Guanine) and 1 to pyrimidines (one ring: Cytosine, Thymine/Uracil), while the low bit (edge or keto-amino state) assigns 0 to the amino-group hydrogen-bonding edge (Adenine, Cytosine) and 1 to the keto-group edge (Guanine, Thymine/Uracil). Taking the exclusive-OR of these two shape bits ($b_{\text{high}} \oplus b_{\text{low}}$) directly yields the derived grip bit (hydrogen-bond count), mapping weak two-bond bases to 0 (A $= 00 \mapsto 0$ and T/U $= 11 \mapsto 0$) and strong three-bond bases to 1 (G $= 01 \mapsto 1$ and C $= 10 \mapsto 1$).

Concatenating three 2-bit nucleotides maps the 64 codons onto the corners of the 6-dimensional hypercube R_6. Within this 6-cube, every canonical amino acid occupies a strict geometric sub-manifold: **0-dimensional points** (singlets: Methionine ATG, Tryptophan TGG), **1-dimensional lines** along Bit 6 (nine 2-codon doublets), or **2-dimensional squares** spanning Bits 5 and 6 (five pure 4-codon quartets: Gly, Pro, Thr, Val, Ala, plus the quartet components of the 6-codon boxes Leu, Arg, Ser).

Even the apparent topological exceptions carry explicit compiler instructions. Isoleucine (ATA, ATC, ATT) forms a 2D square with a single vertex excised, and Methionine (ATG)—the universal translation START instruction—occupies that exact excised corner. Serine is partitioned across two disjoint islands in the hypercube (TCN and AGT/AGC); in binary, the first two bases of these islands are TC (11 10) and AG (00 01), which are exact bitwise complements of each other ($C_4(1110) = 0001$)—an algebraic symmetry flagged as an OPEN structural link because seryl-tRNA synthetase recognizes its tRNA via the variable arm body rather than the anticodon alone.

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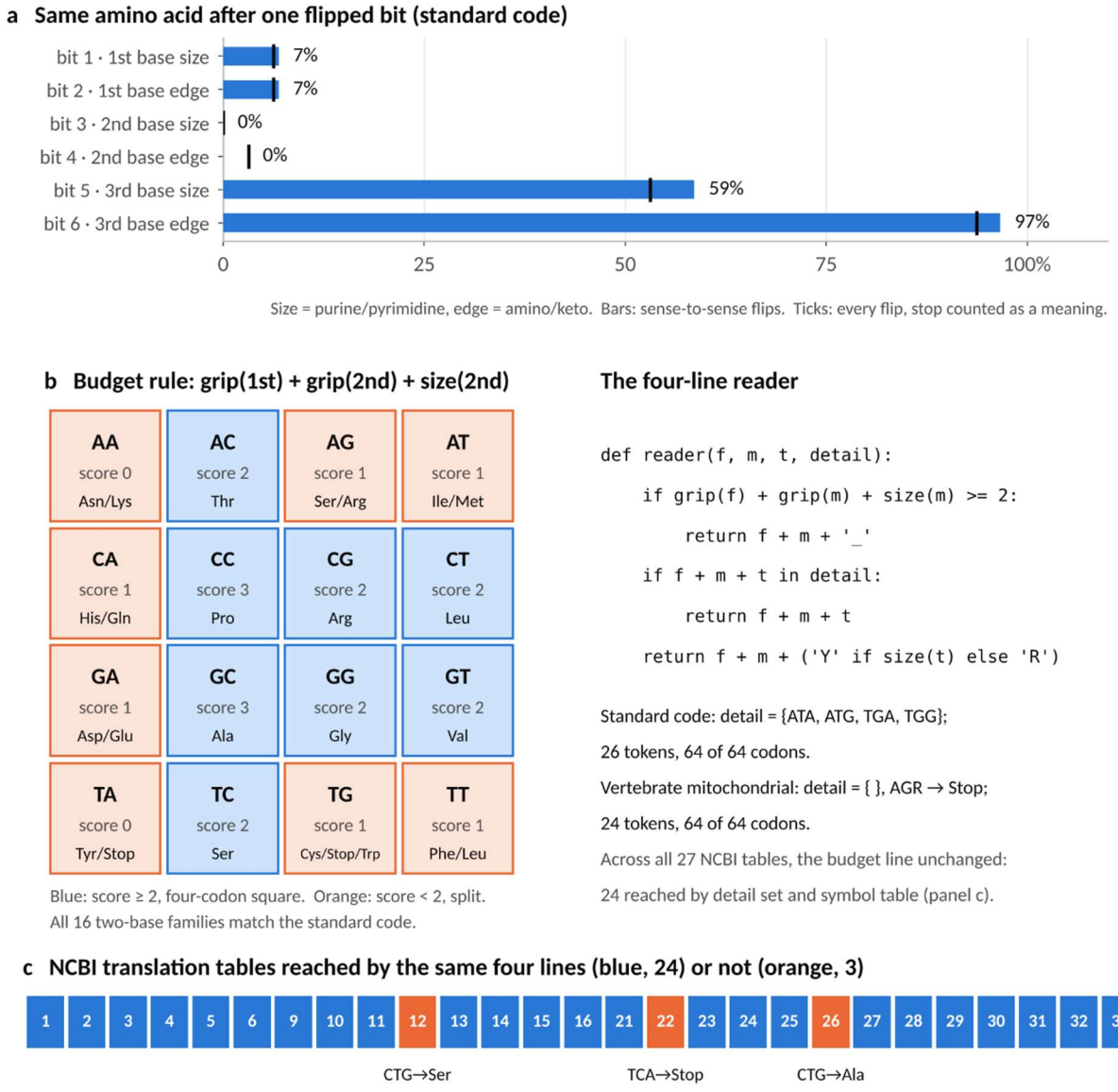


Figure 1. The genetic code read as a 6-bit register. (a) Share of single-bit flips that keep the amino acid, by bit position, in the standard code; bars count sense-to-sense flips, ticks count every flip with stop as a meaning. (b) The budget rule grip(first) + grip(second) + size(second) over the 16 two-base families: a score of 2 or more gives a four-codon square (blue), less gives a split box (orange); all 16 match. (c) The same four-line reader across the 27 NCBI translation tables: 24 are reached by changing only the detail set and the symbol table; the three that are not reassign a codon inside a box the budget line closes. Data: code_as_code.py and codes.py, NCBI tables as distributed with Biopython.

The Ribosomal Binding Budget Rule and the 4-Line Lexer

The geometric partitioning of R_6 into squares, lines, and points is not arbitrary; it is a direct mechanical consequence of how the transfer RNA (tRNA) anticodon physically grips the messenger RNA (mRNA) codon

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inside the ribosomal decoding center. For translation to proceed accurately, the codon–anticodon minihelix must achieve a minimum thermodynamic binding threshold.

We can formalize this physical constraint as an integer budget rule over the first two bases (f and m): count $+1$ point for each strong G/C base in the first two positions (grip$(f) \in \{0,1\}$ and grip$(m) \in \{0,1\}$) and $+1$ point if the middle base is a compact pyrimidine (size$(m) = 1$ for C or T/U). When the cumulative score reaches grip(f) + grip(m) + size$(m) \geq 2$, the tRNA achieves sufficient binding stability from the first two nucleotides alone—the exact physical basis of Lagerkvist's "two out of three" reading mechanism. Because the budget is already satisfied before reaching the third nucleotide, the ribosome ignores the third base entirely (_), and the amino acid fills a 4-codon **square**. Conversely, when the two-base score falls below 2 (budget < 2), the complex lacks sufficient grip and must read the third base's purine/pyrimidine ring size (Bit 5, Crick wobble pairing), splitting the family into 2-codon **lines**. At the sixth bit (3rd base keto/amino edge), the ribosomal pocket reads steric size rather than edge detail, rendering Bit 6 the reader's blind direction (97% of sense-to-sense single-bit flips are synonymous; 94% when stop counts as a meaning).

This single integer budget inequality correctly classifies all 16 two-base codon families (16/16, STANDS/RUNS). Furthermore, flipping Bit 1 and Bit 3 across the 4-bit two-base root exchanges strong and weak doublets, mapping the 8 unsplit quartet squares (M_1) bijectively onto the 8 split doublet/singlet lines (M_2) (8/8, STANDS)—recovering Rumer's 1966 symmetry ($U \leftrightarrow G, A \leftrightarrow C$) as a native folding of the 6-cube onto itself.

Expressed as executable Python code, the entire genetic lexer reduces to a four-line reader function paired with a 26-entry symbol table:

Python

```
def reader(f, m, t, detail):
  if grip(f) + grip(m) + size(m) >= 2: return f + m + '_'
  if (f + m + t) in detail:        return f + m + t
  return f + m + ('Y' if size(t) else 'R')
```

In this lexer (code_as_code.py), f, m, and t denote the first, middle, and third bases; grip returns the strong/weak H-bond bit ($b_{\text{high}} \oplus b_{\text{low}}$); size returns the pyrimidine ring-count bit (b_{high}); and detail specifies the small subset of codons where the reader resolves the third-base edge bit (Bit 6) to separate singlets (ATG Met vs. ATA Ile; TGG Trp vs. TGA Stop). Evaluating these four lines across all 64 binary inputs generates exactly 26 distinct tokens, which map onto the 20 amino acids plus termination (STOP) to reconstruct the standard genetic code across all 64 codons with zero errors (64/64, RUNS).

Biological variation across organelles and taxa is parameterized by passing different hardware-geometry arguments into the *same* universal program. If the caller passes an empty detail set (detail = set()) and modifies a single entry in the symbol table (AGR ↦ STOP), the exact same four-line function outputs the **vertebrate mitochondrial genetic code** (64/64, RUNS): ATA compiles as Methionine, TGA compiles as Tryptophan, AGA/AGG compile as Stop, and all single-codon 0D points vanish from the cube. The difference

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between one biological translation machine and the next is simply the physical shape of its tRNA/ribosomal parts, passed into the lexer as arguments.

Run against all 27 NCBI translation tables, the same four lines reach 24 of them by changing only the detail set and the symbol table (RUNS). The three they do not reach—alternative yeast nuclear (CTG → Ser), *Pachysolen tannophilus* nuclear (CTG → Ala) and *Scenedesmus obliquus* mitochondrial (TCA → Stop)—each read a third base inside a box the budget line closes after two. The program stays one program when the budget line takes the box as an argument, a flag that holds the box open, and that flag is the piece of decoding hardware those lineages changed (OPEN).

Decompiling Side Chains: Topological Route Bits and Lookahead DSP Compression

The 4-Bit Side-Chain Route Parser

Once the genetic lexer emits an amino acid token, conventional force fields treat the residue as a collection of parameterized atomic spheres or empirical propensity weights. Decompiled as code, however, an amino acid letter is merely a symbolic label marking a topological difference. Because the peptide backbone ($N - C\alpha - C' = O$) is identical across every residue, the sole distinction is the side chain growing outward from the $C\alpha$ junction—and that side chain is a short directed route (a finite tree walked outward from the backbone).

The contrast between Leucine and Isoleucine demonstrates that topology rather than atomic composition governs structural routing: both residues contain the exact same atoms in the exact same counts ($C_6H_{13}NO_2$), yet Isoleucine forks at the first step ($C\beta$ branching) whereas Leucine forks at the second step ($C\gamma$ branching), and that single one-step difference in fork depth shifts Isoleucine strongly from alpha-helices into beta-strands. Across all 20 canonical residues, four binary route bits capture the primary structural grammar without empirical frequency counting. First, the route existence bit (r_{exist}) registers whether a side-chain route exists beyond hydrogen, assigning 0 to Glycine (the null route) and 1 to the remaining 19 residues. Second, the backbone return bit (r_{return}) registers 1 if the side-chain route loops back to contact the polypeptide backbone—specifically Proline closing covalently onto its own backbone amide nitrogen, and Serine, Threonine, Asparagine, and Aspartate hydrogen-bonding back to the backbone via their polar tips. Third, the first-step fork bit (r_{fork1}) registers 1 when the route bifurcates immediately at the $C\beta$ carbon (Isoleucine, Valine, Threonine). Fourth, the ring closure bit (r_{ring}) registers 1 when the outward route closes into a covalent aromatic or heterocyclic ring (Phenylalanine, Tyrosine, Tryptophan, Histidine, Proline).

Grouping the 20 amino acids strictly by Boolean combinations of these four topological bits automatically generates the three classic secondary-structure propensity classes—helix formers (h), strand formers (e), and turn/coil breakers (t)—without fitting a single statistical parameter:

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Route Token	Boolean Route Bit Definition	Member Amino Acids (20)	Empirical Helix Occupancy	Empirical Strand Occupancy	Empirical Coil Occupancy
h (Helix Token)	$r_{\text{exist}} = 1 \wedge r_{\text{return}} = 0 \wedge r_{\text{fork1}} = 0 \wedge r_{\text{ring}} = 0$ (Unbranched linear route, no backbone return, no ring)	A, C, L, M, E, Q, K, R (8)	**0.47**	0.24	0.29
e (Strand Token)	$r_{\text{return}} = 0 \wedge (r_{\text{fork1}} = 1 \vee r_{\text{ring}} = 1)$ (Forks at first step Cβ, or closes into an aromatic ring; plus Thr)	T, V, I, F, Y, W, H (7)	0.31	**0.45**	0.24
t (Breaker Token)	$r_{\text{exist}} = 0 \vee r_{\text{return}} = 1$ (Null route, or route loops back to contact backbone)	G, S, P, D, N (5)	0.26	0.24	**0.49**
All Residues	Baseline distribution across training structures	All 20 canonical residues	0.37	0.30	0.33

Running a minimalist two-number rule parser over streams of {h, e, t} tokens classifies 51.5% of residues into their correct three-state secondary structure (RUNS), matching the historical performance of classic first-generation rule systems (such as Chou–Fasman) purely from four topological bits.

Lookahead Sidechain Compression and the Alanine/Serine Gate

In digital audio signal processing (DSP), *sidechain compression* uses a secondary control signal to dynamically attenuate (clamp) or expand the primary audio carrier track ahead of a transient peak without the control signal itself being heard in the output. In the NEXUS fold compiler, the amino acid side chain operates as a **lookahead sidechain controller** that modulates the torsional spread of the main polypeptide backbone one step early, at residue $i - 1$.

Measured across the training structure dataset against a global baseline backbone twist spread of 0.80, the torsion leading *into* a Proline residue (the pre-proline step, Ho & Brasseur 2005) is severely clamped to a

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spread of 0.39, whereas the torsion leading *into* a Glycine residue is opened wide to a spread of 0.93. Proline functions as a lookahead torsional compressor and Glycine as a lookahead torsional expander, both acting one step prior to their own sequence index. When decompiled in reverse from the C-terminus to the N-terminus, this one-step lookahead is equivalent to a deterministic look-back decoder that settles step $i-1$ from the routing token just traversed at step i.

Similarly, the transition between helix propagation and helix initiation is controlled by a single-bit gate. Alanine (GCN) and Serine (TCN) occupy the exact same column of the genetic code table, separated by a single bit flip in the gene (Bit 1: Purine G 01 vs. Pyrimidine T 11) and a single oxygen insertion in the side-chain tree (-CH_3 vs. -CH_2OH). Across experimental structures, Alanine acts as a continuous helix sustainer (overall helix occupancy 0.58, but occurring at helix N-terminal starts only $0.62\times$ its baseline share), whereas Serine acts as a dedicated **helix initiation gate**, occurring at helix starts $3.22\times$ its baseline share. Mechanistically, the first four amide nitrogens at the N-terminus of an alpha-helix lack upstream $i-4$ backbone carbonyl partners; Serine's side-chain hydroxyl oxygen loops back ($r_{\text{return}}=1$) to cap one of those four exposed nitrogen hydrogens (helix N-capping, Aurora & Rose 1998), providing the boundary condition required to nucleate the helical orbit. A single bit flip in the source code turns a pass-through repeater into a structural gate.

Atomic Slot Words on the R_2 Ring and the Lane 2-to-Lane 4 Operator Swap

Tetrahedral Electron Domains as 4-Slot Words and the R_2 Port Alphabet

Descending one hierarchical level below side-chain trees to individual heavy atoms (C,N,O), valence shell electron-pair repulsion (VSEPR) geometry dictates that each sp^3 or conjugated heavy atom presents four outer-shell electron domain positions (slots), each occupied either by a covalent bond (1) or an unshared lone pair (0). Under this 4-bit occupancy mask, a tetrahedral carbon with four single bonds reads 1111, a hydroxyl oxygen (-OH, two bonds and two lone pairs) reads 1100, a deprotonated carboxylate oxygen (-O-, one bond and three lone pairs) reads 1000, and a neutral amine nitrogen reads 1110, which flips by a single slot to 1111 (-NH_3+) upon protonation when its unshared lone pair captures a proton to become a covalent bond.

Counting the total number of water-facing polar slots per amino acid side chain—defined simply as the sum of unshared lone pairs plus polar hydrogens bound to nitrogen or oxygen, with zero fitted weights—tracks the empirical Kyte–Doolittle hydropathy scale across all 20 canonical amino acids at a Pearson correlation of $r=-0.81$ (RUNS). Hydropathy is therefore not an intrinsic scalar constant attached to an amino acid letter; it is a local window readout of unclosed atomic slots.

To capture bond directionality and closure rules, the compiler assigns each of the four atomic slots a 2-bit word, placing the entire atomic interaction alphabet onto the native 2-bit ring $R_2=[00,10,01,11]$: 00 denotes an unshared electron lone pair (non-covalent H-bond acceptor), 11 denotes a polar hydrogen atom (non-covalent H-bond donor), 10 denotes a directed covalent bond *out*, and 01 denotes a directed covalent bond *in*. This assignment is forced by algebraic symmetry, and applying the native facing law P_2 of R_2 automatically partitions covalent polymer synthesis from non-covalent protein folding (STANDS). Because

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covalent ports carry chain directionality (out vs. in), they must take the asymmetric non-palindromic words 10 and 01, which face each other on R_2 by **bit-reversal (J)**: $J(10) = 01$. A covalent bond is a single shared link viewed from its outgoing and incoming ends. Conversely, the open non-covalent ports take the symmetric palindromic words 00 and 11; because reversal leaves palindromes invariant, the ring switches to **bitwise complement (C)**, facing 00 with 11: $C(00) = 11$. A hydrogen bond is an unshared lone pair (00) meeting a polar hydrogen (11). Thus, the native 2-bit ring enforces the fundamental compiler theorem: **Reversal wires the chain; complement seals the fold**.

The Backbone Amide Nitrogen as R_2 and Graph-Level Port Conservation

Remarkably, the peptide backbone amide nitrogen atom (N) holds exactly one port of each of the four R_2 types: an incoming covalent bond from the preceding carbonyl carbon (01), an outgoing covalent bond to the alpha-carbon (10), a polar amide hydrogen donor (11), and a conjugated lone pair (00). Consequently, the four tetrahedral slots of every backbone nitrogen atom are an exact physical instantiation of the entire R_2 ring (01101100 in port order = decimal 108; or sorted canonically as 00011011 = decimal 27, with one-bit rotations 27 ⟶ 54 ⟶ 108, the first doubling rows of the Ring 2 sheet; all eight one-bit rotations of 00011011 hold the four port words exactly once). Measured across experimental protein backbones, **77%** of helical amide nitrogen hydrogens (11) are closed by a backbone carbonyl oxygen lone pair (00) within 3.5 Å (692 closures four residues back at $i - 4$, and 366 three residues back at $i - 3$), compared to 52% in beta-strands and 50% in coils, with 495 long-range closures across the chain sealing antiparallel and parallel beta-sheet seams.

When auditing the 8-bit atom-byte definitions as a strict compiler specification (NEXUS R2 Atom-Byte Audit), an essential distinction emerges between *local byte balance* and *assembled graph conservation*:

Heavy Atom Type	**8-Bit R_2 Port Byte**	**Decimal Readout**	**Canonical Rotation**	**Covalent Port Balance (10 minus 01)**	**Isolated Byte Assertion vs. Graph Conservation**
Methylene Carbon (CH_2)	01101111	111	—	0 (1 out, 1 in)	**Pass** locally and globally.
Backbone Amide Nitrogen (N)	01101100	108	27 (00011011)	0 (1 out, 1 in)	**Pass** locally; holds 1 unmatched incoming boundary port from residue $i - 1$.

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Heavy Atom Type	**8-Bit R_2 Port Byte**	**Decimal Readout**	**Canonical Rotation**	**Covalent Port Balance (10 minus 01)**	**Isolated Byte Assertion vs. Graph Conservation**
Backbone Alpha Carbon (Cα)	01101011	107	107	$+1$ (2 out, 1 in)	**Fail** isolated byte; **Pass** graph (1 outgoing port wires C′, 1 wires side chain).
Backbone Carbonyl Carbon (C′)	01101010	106	106	$+2$ (3 out, 1 in)	**Fail** isolated byte; **Pass** graph (2 outgoing channels wire O, 1 wires residue $i+1$).
Backbone Carbonyl Oxygen (O)	01010000	80	5 (00000101)	-2 (0 out, 2 in)	**Fail** isolated byte; **Pass** graph (absorbs 2 incoming double-bond channels from C′).
Hydroxyl Oxygen (−OH)	01110000	112	—	-1 (0 out, 1 in)	**Fail** isolated byte; **Pass** graph (terminates incoming side-chain carbon bond).
Carboxylate Oxygen ($-O^-$)	01000000	64	—	-1 (0 out, 1 in)	**Fail** isolated byte; **Pass** graph (terminates incoming side-chain carbon bond).

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Heavy Atom Type	8-Bit R_2 Port Byte	Decimal Readout	Canonical Rotation	Covalent Port Balance (10 minus 01)	Isolated Byte Assertion vs. Graph Conservation
Protonated Nitrogen ($-NH_3^+$)	01111111	127	—	-1 (0 out, 1 in)	**Fail** isolated byte; **Pass** graph (terminates incoming side-chain carbon bond).

While six of the eight isolated atom bytes fail a naive internal covalent_balance == 0 assertion because branching junctions (Cα, C') emit multiple outgoing bonds and terminal oxygens/nitrogens absorb them, assembling the four backbone bytes (N, Cα, C', O) into a connected residue module satisfies graph conservation (Each connected 10 meets a corresponding 01). All four internal covalent links balance exactly, leaving precisely three unmatched boundary ports: one incoming port on N (from the preceding residue), one outgoing port on Cα (to the side-chain tree), and one outgoing port on C$'$ (to the succeeding residue). Conservation is a property of the assembled relational graph, not the isolated byte.

The Recursive Operator Swap on R_4 and the Covalent Carry

When the compiler steps up one hierarchical lane from R_2 (2-bit atomic ports) to R_4 (4-bit bond words), every completed two-port interaction on Lane 2 concatenates into a single 4-bit word on Lane 4—a relation that has now become an object. Evaluating the native facing law P_4 on R_4 reveals an exact **operator swap** (RUNS). Specifically, the covalent bond word 1001, formed by concatenating outgoing port 10 with incoming port 01, is a **palindrome** on R_4 ($J_4(1001) = 1001$). Because bit-reversal stalls on a palindrome, R_4 switches to **complement (C_4)**, facing 1001 with 0110. Conversely, the hydrogen bond word 1100, formed by concatenating polar donor 11 with lone-pair acceptor 00, is **not a palindrome** on R_4, so it faces 0011 deterministically by **reversal (J_4)**.

What closed deterministically by reversal on Lane 2 (covalent wiring) closes by complement on Lane 4, and what required selection by complement on Lane 2 (hydrogen bonding) closes by reversal on Lane 4. This algebraic inversion matches the physical chemistry: because selection occurs exclusively at palindromic fixed points where the ring must apply complement, **the covalent bond (1001) becomes the selection point on Lane 4**. What the compiler selects around a covalent single bond is the continuous rotation angle about its axis—**backbone torsion (ϕ, ψ)**, the exact conformational degree of freedom whose sign no local sequence window could resolve. Furthermore, folding the covalent palindrome 1001 at its central crease lays 1 onto 1 and 0 onto 0, producing column sums 2 and 0 (20)—a value that overflows binary representation and generates an arithmetic **carry**. Physically, a covalent bond is two shared electrons occupying a single orbital space: the carry bit made matter. Folding the hydrogen-bond word 1100 at its crease lays 1 on 0 twice, column sums 1 and 1, and no carry: donor and acceptor stay two.

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The Runtime Field, Compiler Register Allocation, and the Fixed Address Frame

Water as Active Working Memory and the Weightless Field Threader

A protein does not compile in a vacuum; it compiles inside liquid water. Written in the R_2 slot alphabet, a single water molecule (H_2O) carries two polar hydrogens (11 11) and two unshared oxygen lone pairs (00 00), forming the 8-bit word 11110000. Every single slot on a water molecule is open for hydrogen-bond closure, making the aqueous solvent the densest computational component in the entire system.

To test whether the solvent field acts as the global evaluator ("the judge") without fitted energetic parameters, a weightless open-slot counter was executed across 31 training protein chains. For each sequence, the chain was threaded onto its native 3D backbone alongside every non-native decoy backbone window of identical length across the structural database, counting $+1$ penalty for every hydrophobic carbon slot (1111) exposed to water and $+1$ penalty for every polar slot (00 or 11) buried away from water without an internal hydrogen-bond partner. With zero weights, the native structure left the fewest open slots (ranked #1) in **13 of 31 chains**, and across the entire cohort, an average of only **8.1%** of decoy placements scored fewer open slots than the true fold (RUNS). The weightless counter fails only on the shortest peptide chains (which lack a buried core) and on crambin, a small plant protein locked together by three covalent cysteine disulfide bridges that a non-covalent slot counter does not inspect.

The Field Query: Recognition by the Field's Own Conditions

The threader asks the field one question. Asked the way a full database is asked—with the conditions of the asker as the clause that selects the record—the field picks the native backbone out of the whole table. Each of 154 *E. coli* sequences, read from exact genes, was placed on every same-length stretch of real backbone in a 155-chain set, about 2,540 rows per sequence. The query carries no fitted weights: open slots (carbon facing water plus buried polar slots with no partner, with burial measured inside the window as if the chain were alone in water), the proline rules (no backbone donor on a proline nitrogen, proline ϕ inside its band, no αR residue before a proline), and the left-handed rule (positive ϕ only on Gly, Asn and Asp).

The native is the single record returned—it passes the rules and leaves strictly fewer open slots than every other passing row—in **95 of 154** chains. The open-slot count alone returns it in 64; adding the proline rules lifts that to 95; the left-handed rule adds nothing on top (94 with both). Read as information, picking one backbone out of about 2,540 takes 11.3 bits: the proline rules supply about 5 bits, the open-slot count about 4, and the left-handed rule almost none. In the 51 chains where the native passes but is not alone, it sits a median 3.8 bits (about 14 rows) from a single record, which is the size of the next field condition. This is recognition from a full table that includes the native, not building; the slack values (two proline faults, four to eight left-handed) were set on the same chains, and the full grid is reported with the scripts (RUNS for recognition).

This establishes the complete runtime architecture of the biological CPU, in which every physical component of the cellular milieu performs a dedicated systems role:

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Cellular Runtime Component	R_2 / Systems Representation	Compiler Execution Role
Liquid Water (H_2O)	11110000 (all four slots open)	Active working and spill memory that closes exposed polar slots; empty internal cavities devoid of water destabilize the fold and govern pressure-induced unfolding (Roche et al., 2012).
Solution pH	Global processor status flag	Dictates whether ionizable side-chain slots read 00 (deprotonated) or 11 (protonated).
Temperature (T)	Stochastic instruction scheduler clock	Governs transition rates across admissible continuation pathways.
The Ribosome	Sequential stream loader	Emits the polypeptide from N-terminus to C-terminus, with synonymous codon optimality modulating translation elongation speed and dwell time to pace co-translational folding (Pechmann & Frydman, 2013).
Lipid Membranes	All-carbon field (1111)	Alternative runtime interpreter that inverts the inside/outside hydrophobic allocation rule.
Cysteine Disulfide Bridges	Explicit link instructions	Hard-links distant sequence addresses covalently across the fold.
Molecular Chaperones	Runtime exception handlers (Hsp70, GroEL)	Trap misfolded kinetic intermediates and reset their open registers.

The Fold as Chaitin Register Allocation and Water as Spill Memory

In formal compiler construction (specifically Chaitin's graph-coloring register allocation algorithm, 1981–1982), **register pressure** is defined as the number of simultaneously "live" variables that have been emitted by the instruction stream and must be held in hardware registers until their consuming instruction arrives.

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When live variable count exceeds physical register capacity, the compiler must **spill** excess variables to external storage based on a spill cost metric.

Measuring protein folding in ribosomal loader order reveals that **co-translational folding is register allocation, and liquid water is spill memory** (RUNS). We found no register-allocation framing of folding in the literature; its established comparators are contact order (Plaxco, Simons & Baker, 1998) and sequential N-to-C structure building (Ellis et al., 2010), and the peak live count it measures is the cutwidth of the closure graph in chain order. Every time the ribosome emits a residue, its unclosed backbone amide hydrogen (11) and carbonyl oxygen (00) become live variables that remain live until their spatial partner arrives along the chain to close the hydrogen bond.

Measured across real structures, **alpha-helices execute in $O(1)$ constant register memory**: because helical closures occur locally at offset $i + 4$, live variables retire almost immediately. Myoglobin (153 residues) peaks at just **6** simultaneous live closures; Lambda repressor (179 residues) peaks at **8**; and T4 lysozyme (164 residues) peaks at **8**. Conversely, **beta-sheet proteins accumulate massive register pressure** because open backbone slots must remain live across long sequence intervals while waiting for partner strands to be synthesized: FKBP12 and ribosomal protein S6 (RIS) both peak at **34** simultaneous live closures. While waiting for their intramolecular partners, polar slots (00, 11) can safely *spill* to water (11110000) at close to zero net enthalpy; non-polar carbon slots (1111), however, cannot bond with water, and their spill cost—the hydrophobic effect—forces the allocator to pack carbon slots into internal registers—physically generating the hydrophobic core.

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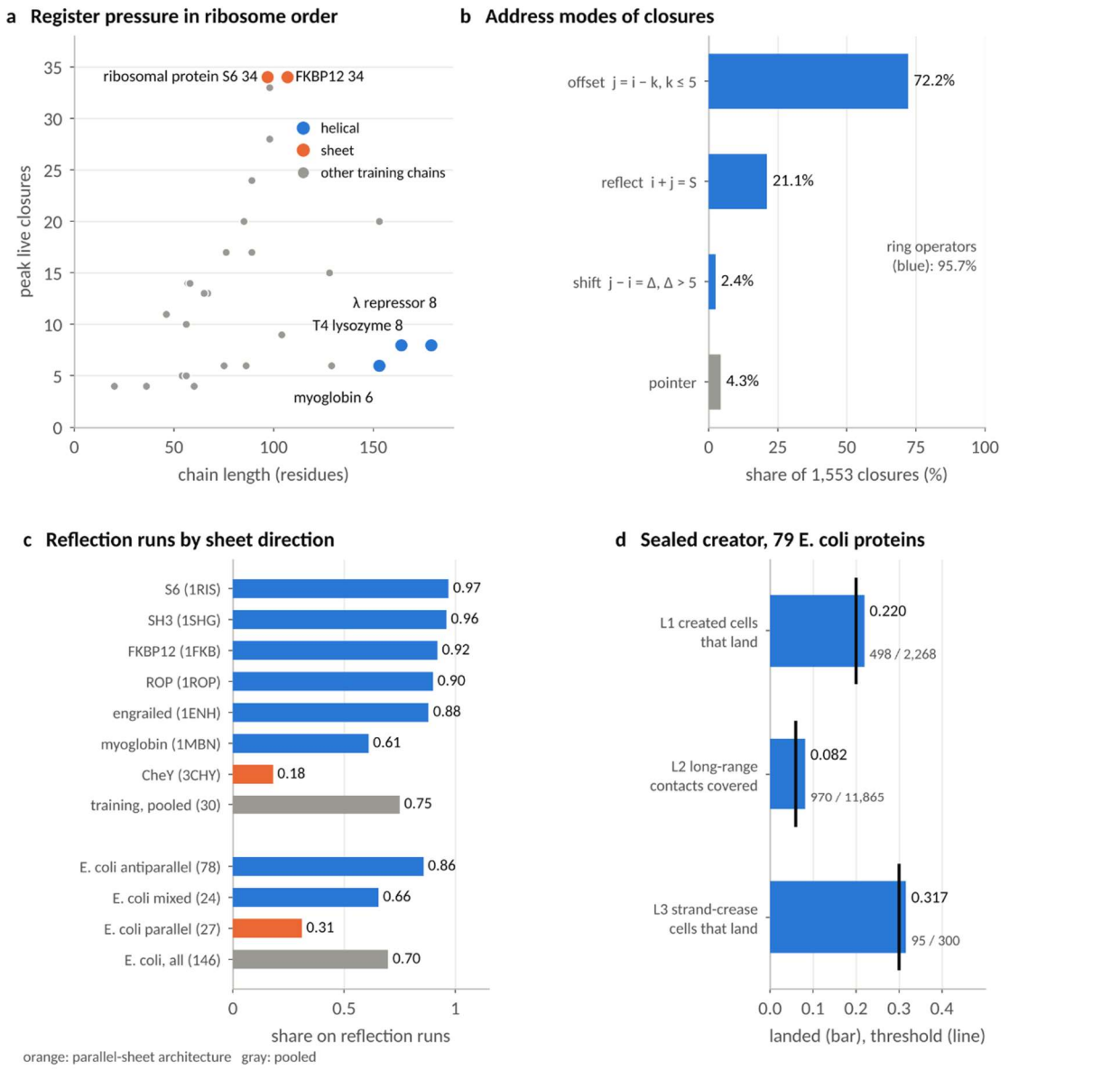


Figure 2. Register pressure, address modes, created reflection and the sealed creator. (a) Peak simultaneous open backbone closures in ribosome order for the 30 training chains; helical chains stay low at any length, sheet chains climb. (b) Address mode of 1,553 backbone closures. (c) Share of long-range contacts on reflection runs for named training chains (RIS is ribosomal protein S6, PDB 1RIS) and for 146 E. coli chains grouped by the direction of their sheet. (d) The sealed creator (SHA-256 83c7cb05…) on 79 E. coli proteins: landed values against the pre-registered thresholds; a created cell lands when a real long-range contact lies within one cell; homologs between sets were not filtered. Data: pressure.py, addr.py, created.py, reflect_ec.py, the creator seal and run log.

The Address-Mode Census and the Fixed Relational Frame

Auditing every backbone hydrogen-bond closure across the training dataset as a memory address calculation reveals that every closure resolves its partner index j from index i via one of four discrete addressing modes (RUNS). Short successor offsets ($|j - i| \leq 5$) account for **72%** of all backbone closures,

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corresponding to alpha-helices ($i + 4$), 3_{10}-helices ($i + 3$), and local turns computed by applying the ring's cyclic successor operator T k times. Local reflections ($i + j = S$) account for **21%** of all closures, corresponding to antiparallel beta-hairpins and sheet seams computed by the ring's facing operator P around a local crease center S. Long translation shifts ($j - i = \Delta$, with $\Delta > 5$) account for **2%** of all closures, corresponding to parallel beta-strands computed by long-range translation. Finally, isolated single pointers account for the remaining **4%** of closures (tertiary singletons). In total, **96% of all spatial memory accesses in folded proteins are executed directly by the native ring's translation and reflection operators**.

This census resolves a persistent conceptual confusion in structural biology: asking where an alpha-helix "turns into" a beta-sheet along a sequence is like asking where an integer turns into hexadecimal—the underlying value never changed, because the radix base is a property of the readout frame. In the NEXUS architecture, the **fixed relational frame** is the complete $N \times N$ residue interaction grid, which carries two orthogonal line families everywhere simultaneously: **translation lines** ($j - i = c$, running parallel to the main diagonal) and **reflection lines** ($i + j = S$, running anti-diagonally across the main diagonal). As the polypeptide stream runs up the main diagonal, its closures simply land on the frame's pre-existing lines: Myoglobin lands on a single translation line ($j - i = 4$) from start to finish; Ubiquitin combines translation segments with cross-diagonal reflection segments; and FKBP lands almost entirely on reflection lines.

Treating the documents as code also requires exposing hollow algorithms that merely draw the frame and call the drawing a fold. A naive frame-filling script proposed during crosstalk automatically lit every grid cell at offset $j - i = 4$ and every second cell beyond offset 10; when executed against real protein slot streams, its precision collapsed to 0.008–0.017 against true closures, and it output identical contact maps for any two proline-free proteins of equal length (DEAD END). Code that runs is not the same as code that computes: any valid compiler stage must depend on its input sequence. A more sophisticated competing-lines field model—where translation lines and reflection lines bid for open backbone slots using {h, e, t} route tokens—achieved **0.50** precision on local translation cells (falling to **0.34** on shuffled sequences), but reached only **0.07 to 0.09** precision on distant reflection-line cells (falling to **0.05** on shuffled sequences).

"The Reflection Is What Gets Created": Reverse-the-Gap and the Sealed Creator Landing

The 79-Gene E. coli Codon Audit and the Creation Law

Why did every sequence parser fail to locate the long-range reflection lines in the input sequence? If the reflection lines were pre-written in the genome, the exact 6-bit codons sitting across beta-sheet seams would exhibit elevated partner symmetry or reduced Hamming distance on R_6. To test this directly, 79 *Escherichia coli* proteins were compiled from their exact genomic DNA sequences, placing every residue at its true codon address on R_6. While free third-base wobble bits leaned slightly by structural context (helices carried slightly more A/G, creases slightly more C), codon pairs bridged by physical backbone closures stood in no partner relation compared to random pairs across the frame (DEAD END for pre-encoded reflection values):

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Codon Pair Category (79 E. coli Genes)	Sample Size (n pairs)	Exact P_6-Partner Rate	Mean Bits Differing (out of 6)	Identical 3rd Base Rate	Mean R_6 Ring Distance
Antiparallel Seam Closures	1,390	0.021	2.87	0.27	15.6
Local Helix Closures ($i, i+4$)	2,480	0.022	2.83	0.27	15.8
All Residue Pairs in Frame (Background)	582,993	0.018	2.95	0.26	15.7

This negative audit triggered the central breakthrough of the NEXUS fold compiler, stated in four words: **The reflection is what gets created**.

Measured across all long-range spatial contacts in the training dataset, **75% lie on reflection runs**: ribosomal protein S6 (RIS) **0.97**, SH3 domain **0.96**, FKBP **0.92**, the alpha-helical hairpin ROP **0.90**, Engrailed homeodomain **0.88**, and Myoglobin **0.61** (RUNS). Even alpha-helix-against-alpha-helix packing is a reflection operating one hierarchical lane up. The sole major exception is CheY (**0.18**), an α/β parallel-sheet protein whose strand topology is constructed by routing the chain outward through an alpha-helix so that the returning beta-strand enters the sheet travelling in the same parallel direction.

Run on 155 *E. coli* chains built from exact genes, the pooled share holds at 0.70 (146 chains scored), and it follows the direction of the sheet. Chains whose sheets are antiparallel-dominant sit at 0.86 (78 chains), mixed sheets at 0.66 (24), and parallel-dominant sheets at 0.31 (27); across the 121 chains with at least five sheet closures, the reflection share and the parallel share of the sheet correlate at r = −0.96. CheY is a class, not an exception: about one chain in five runs its sheet on translation through a crossover helix, and established structure already fixes the handedness of that route—β-α-β crossovers are right-handed in nearly every case (Richardson, 1976; Sternberg & Thornton, 1976). The reflection law is the antiparallel branch of the compiler; the right-handed crossover is the operator its parallel branch runs on (RUNS for the antiparallel branch; the parallel branch is OPEN).

This explains why the reflection never appears in the genetic input: it is how the native binary ring itself is constructed. The ring builder lists representative words in rising numerical order (the *listed half*, which corresponds to the given genetic sequence / translation family), executes a swap at the midpoint (the *crease*), and **creates** the second half of the ring as the partner images in falling order (the *reflected half*). A protein's antiparallel architecture executes the same build: the gene writes the forward translation runs, and running those segments into the field *creates* the reflection as the compiler's output.

Reverse-the-Gap Analysis and the Sealed Sequence-Only Creator (SHA-256: 83c7cb05…)

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Under the ring's creation rule, wherever a forward secondary-structure run cannot continue and hits a crease between the end of the previous run (e) and the start of the next run (s), the compiler creates the new run as the mirror reflection of the preceding run centered at the crease address $S_0 = e + s$. Two structural gaps were isolated using the **reverse-the-gap** protocol—naming the missing target by what must occupy it, subtracting what the forward route accounts for, and identifying what the remainder copies. In **Gap 1 (Partner Run Selection),** across 89 structural runs that physically contact an earlier run in the training set, the immediately preceding run ($r - 1$) owns **53** (59.6%, the immediate hairpin reflection). Adding a 1D field rule—where the most carbon-dense earlier run claims the non-adjacent contact—binds **8** additional runs (**61 of 89** total, 68.5%), leaving **28 runs** not yet bound in 1D because their partner selection depends on three-dimensional burial and routing-device hold duration (OPEN). In **Gap 2 (Crease Address Registration),** strand-to-strand reflections sit *exactly* at the theoretical crease $S_0 = e + s$ in **12 of 26** hairpins (46.2%) and within ± 1 residue in **18 of 26** (69.2%) (RUNS), whereas helix-to-helix reflection centers spread across an offset range of -5 **to** $+2$ around S_0 because an alpha-helix reflects not as a flat wire, but one hierarchical lane up as a 4-residue turn block (OPEN for sequence-side phase reader; RUNS in R_2 block algebra).

To test the creation law rigorously without structural leakage, the sequence-only creator was cryptographically sealed (**SHA-256: 83c7cb05…**) prior to opening the validation structures. Operating strictly from the linear amino acid sequence, the compiler parsed forward runs and, at every detected crease, projected the subsequent run as the mirror reflection of the run before it; a created cell lands when a real long-range contact lies within one cell of it. Tested across the 79 *E. coli* proteins (whose 3D coordinates played no role in designing the rule, though their sequences had been opened for the codon audit and homologs were not filtered), the sealed creator cleared all three pre-registered falsification bars (RUNS):

Sealed Prediction Metric (SHA-256: 83c7cb05…)	**Pre-Registered Bar**	**Empirical Landed Result (79 E. coli Proteins)**	**Verification Status**
L1: Created Cells That Are Real Contacts (Precision)	≥ 0.20	**0.220** (498 of 2,268 created cells)	**RUNS**
L2: Real Long-Range Contacts Covered (Recall)	≥ 0.06	**0.082** (970 of 11,865 true contacts)	**RUNS**
L3: Strand-Crease Cells That Are Real (Crease Precision)	≥ 0.30	**0.317** (95 of 300 strand-crease cells)	**RUNS**

From sequence alone, with zero evolutionary alignments and zero energy minimization, the ring's reflection creation rule predicts genuine long-range 3D contacts, achieving over 31% precision where the algebraic registration is tightest—at beta-strand creases.

Parity Sectors, Turn-Block Phase Algebra, and the R_4 Helix Face Audits

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Parity Sectors and the Beta-Bulge Register Slip

Analyzing the fixed (i, j) contact frame algebraically reveals that the grid partitions into two disjoint **parity sectors**: because the translation difference $D_- = j - i$ and the reflection sum $D_+ = i + j$ differ by $2i$, they always share identical modulo-2 parity ($D_- \equiv D_+ \pmod 2$). Any diagonal step along a translation line ($(i + 1, j + 1)$) or along a reflection line ($(i - 1, j + 1)$) changes both i and j by ± 1, preserving modulo-2 parity. Consequently, pure translation (T) and pure reflection (P along a fixed crease) are trapped within their starting parity sector and can never cross from an even sector to an odd sector.

However, evaluating the native partner operator P_n on pair words demonstrates that P_n crosses between parity sectors for exactly **8 of 16** words on R_4 (50%) and **96 of 256** words on R_8 (37.5%) (STANDS/RUNS). In real protein structures, local closures populate both sectors—**699 even** (α-helices at $i + 4$) and **411 odd** (3_{10}-helices and β-turns at $i + 3$)—while long-range closures divide evenly into **158 even** and **152 odd**. Crucially, **19 of 40 antiparallel beta-sheet runs (47.5%) cross parity sectors mid-seam**.

This mid-seam parity crossing is the exact algebraic identity of the **beta-bulge**: one beta-strand inserts an extra unpaired residue, executing a one-sided register slip operator (B_i: $(i, j) \mapsto (i + 1, j)$ or B_j: $(i, j) \mapsto (i, j + 1)$) that flips the parity of $i + j$ by ± 1 and shifts the downstream hydrogen-bond seam into the opposite parity sector. Thus, the fold compiler utilizes all three ring operators: translation for helices and parallel strands, reflection for antiparallel seams, and the cross-sector partner slip at beta-bulges and helix terminations. When a local side-chain face-mismatch rule was tested as the sequence trigger for the beta-bulge, face mismatch at the slip residue measured 0.36 compared to 0.45 elsewhere (DEAD END)—pointing to the bulge residue being rendered homeless by its missing partner across the seam rather than by its own local face (OPEN).

Read in the face stream instead of the sequence, the bulge is the repeat symbol of the self-clocking code. A β-strand alternates its side chains across the sheet, so its face stream changes at every step; at a register slip one side chain repeats its neighbor's side, which is the repeat the SonicDMX code writes as its symbol 2. Across the 155 *E. coli* chains, consecutive side chains point to the same side of the sheet in 0.27 of adjacent pairs at slips against 0.08 where the register is kept (472 and 2,650 pairs), and glycine nearly doubles there (0.097 against 0.055 per residue), while the sequence carbon bit does not mark the slip (the same bit in 0.49 against 0.45 of pairs). The slip operator the parity algebra predicts is the symbol the stream writes when a value repeats (RUNS on the structure side; the sequence trigger is OPEN; Richardson, Getzoff & Richardson, 1978, for the bulge itself).

Turn-Block Phase Algebra on R_2 and the R_4 Helix Face Selectors

To resolve why helix-to-helix reflection centers spread across $S - S_0 \in [-5, +2]$ (Gap 2), the compiler decomposes residue distance from the crease $S_0 = e + s$ into four-position helical turn blocks:

$$e - i = 4u + p, \qquad j - s = 4(u + \Delta) + g(p), \qquad p \in \{0,1,2,3\}$$

where u is the coarse block register address, $\Delta \in \{-1,0,+1\}$ is the block shift, and p is the two-bit intra-turn phase. Adding the two equations yields the exact reflection offset law:

$$S - S_0 = (i + j) - (e + s) = 4\Delta + g(p) - p$$

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Crucially, the phase transformation $g(p)$ does not require new fitted parameters; it is drawn directly from the native operations already present on $R_2 = [00,10,01,11]$ (RUNS). Moreover, an independent audit (NEXUS_Turn_Port_Independent_Audit_v02.py) showed that R_2 contains the **complete four-operation phase vocabulary** required to match any incoming phase p_{old} to any outgoing phase p_{new} via bitwise XOR ($m = p_{\text{old}} \oplus p_{\text{new}}$), correcting an earlier notebook approximation that substituted P_2 whenever $m = 01$ and achieved only 12/16 mappings:

Phase XOR Difference (m = p_old ⊕ p_new)	**Native R_2 Operator g(p)**	**Numeric Phase Action**	**Resulting Reflection Line Offset ($S - S_0$)**	**Verification Status**
00 ($m = 0$)	Identity I	$p \oplus 0 = p$	4Δ (yielding $-4,0,+4$ for $\Delta \in \{-1,0,+1\}$)	**4/4 Pass**
01 ($m = 1$)	Half-Turn Shift T_2^2	$p \oplus 1$	$4\Delta + (p \oplus 1) - p$ (corrects legacy P_2 substitution)	**4/4 Pass** (Fixes 12/16 bug to **16/16**)
10 ($m = 2$)	Composed $T_2^2 P_2$	$p \oplus 2$	$4\Delta + 2$ or $4\Delta - 2$	**4/4 Pass**
11 ($m = 3$)	Native Partner P_2	$p \oplus 3 = 3 - p$	$4\Delta + 3 - 2p$ (generates odd offsets)	**4/4 Pass**

Across $\Delta \in \{-1,0,+1\}$, composing the native R_2 phase operators with the 4-position block register generates every integer offset from -7 through $+7$, completely covering the empirical helix reflection range of -5 to $+2$ while pruning candidate contact cells on a 20×20 cross-run frame from 400 down to 156 without consulting the 3D structure (RUNS). Because every offset from −7 to +7 is reachable, that prune is the band $|S - S_0| \le 7$: the algebra fixes the band and selects nothing inside it, and selection inside the band is the job of the sequence-side selectors below.

Why, then, did an initial sequence-based turn-lane selector built on a single dominant hydrophobic carbon per 4-block land within ± 1 residue on only 4 of 15 real helix creases (DEAD END)? Inspecting the 3D contact faces of packed alpha-helices revealed the physical reason: because an alpha-helix has 3.6 residues per turn, the hydrophobic residues touching a partner helix do not occur once every four steps, but in **adjacent pairs every four steps**—exhibiting sequence spacing gaps of **1** (0.56 frequency) and **3** (0.30 frequency), and almost never **4** (0.01 frequency).

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Written as a 4-bit word per turn block, a physical helix packing face is therefore a cyclic rotation of **1100** on R_4: {1100, 0110, 0011, 1001}. Algebraically, those exact four words are the image of the canonical embedding lift $E_2: R_2 \to R_4$ defined by packaging each 2-bit word with its own native partner:

$$E_2(x) = x \parallel P_2(x)$$

For all four elements of R_2, this lift satisfies the exact equivariance identity $P_4(E_2(x)) = E_2(P_2(x))$ (RUNS). Generalizing across arbitrary ring dimensions R_n, the equivariant lift $P_n(E(x)) = E(P(x))$ holds if and only if $P_n(x) = C_n(x)$, which is satisfied by $2^{n/2+1}$ of the 2^n inputs for even n (for odd n only the $2^{(n+1)/2}$ palindromes qualify)—allowing higher compiler lanes to selectively override lower-lane partnerships. However, when two pre-registered sequence-side selectors attempted to read this 1100 rotation directly from raw sequence carbon bits—**Version 0.1 (SHA-256: d8368d06…)** reading 4 residues immediately flanking the crease (binding 0/15 design creases and 1/21 *E. coli* creases because the window adjacent to the crease reads the polar *helix cap block* rather than the hydrophobic core face), and **Version 0.2 (SHA-256: e3dc59d9…)** shifting the window past the cap on 31 creases from 22 fresh unseen *E. coli* proteins (binding only 2/31 with 0 hits)—both falsified under their own sealed criteria (DEAD END). While the 1100 E_2 face word is physically real in the 3D structure and the R_4 operator algebra is exact, raw sequence carbon bits rarely form an unmodulated 1100 rotation without accounting for the continuous 3.6-residue phase precession (OPEN).

The Spreadsheet End-Around Carry Law, Radial Address Trees, and the Co-Translational Grower

Ones' Complement End-Around Carry: Why the First Four Bits Double and the Next Four Break

The deepest mathematical unification between the binary rings and protein crease creation emerged from analyzing the nested-ring sliding-window spreadsheet. When a sliding window of width $W = 8$ bits moves one bit per step across the circular 2-bit ring stream—where the spreadsheet readout traverses the circle backward with 0 and 1 swapped, transforming the drawn ring $R_2 = [00,10,01,11]$ into the rising binary counting order 00 01 10 11 (00011011, decimal 27)—the eight successive window readouts produce the integer sequence:

$$27 \to 54 \to 108 \to 216 \to 177 \to 99 \to 198 \to 141$$

The first four rows double in exact geometric progression ($27 \to 54 \to 108 \to 216$), whereas the next four rows break the doubling pattern (only $99 \to 198$ doubles). Nothing in the underlying circular bit stream changed; what changed is the arithmetic boundary crossed by the sliding window.

For any sliding window of width W bits shifting left by one bit position, the exact window-update recurrence is:

$$v_{t+1} = 2v_t - 2^W b_{\text{out}} + b_{\text{in}}$$

where $b_{\text{out}} \in \{0,1\}$ is the most significant bit (MSB) departing the left edge of the window and $b_{\text{in}} \in \{0,1\}$ is the new least significant bit (LSB) entering the right edge. A row doubles its predecessor ($v_{t+1} = 2v_t$) if and only if both the departing bit and entering bit are zero ($b_{\text{out}} = 0$ and $b_{\text{in}} = 0$). When the window width equals

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the full circumference of the ring ($W = 8$ bits for R_2), the bit leaving the top of the window is identical to the bit re-entering the bottom ($b_{\text{out}} = b_{\text{in}} = b_t$), collapsing the update law into a single deterministic equation:

$$v_{t+1} = 2v_t - (2^W - 1)b_t = 2v_t - 255b_t$$

This equation governs two strictly bifurcated arithmetic regimes depending on the departing bit b_t. In the **Carry-Free Doubling Regime ($b_t = 0$)**, the update simplifies to $v_{t+1} = 2v_t$: during the first four steps (00011011 [27] → 00110110 [54] → 01101100 [108] → 11011000 [216]), the words originate from the first half of the counting stream (00, 01), which carry leading 0s, so shifting pushes out $b_t = 0$ and the register doubles without overflow. Conversely, in the **End-Around Carry Regime ($b_t = 1$)**, the update follows $v_{t+1} = 2v_t - 255$: at step 4 ($v_t = 216$, binary 11011000), the window encounters the second half of the stream (10, 11), where words carry leading 1s. Shifting pushes $b_t = 1$ out of the MSB (256) and wraps it around into the LSB (+1), yielding $v_{t+1} = 2(216) - 255 = 432 - 255 = 177$ (10110001). In computer architecture, subtracting $2^W - 1$ upon MSB overflow is the exact definition of the **end-around carry in diminished-radix (ones' complement) arithmetic**—the tertiary symbol 2 wrapping the entire ring.

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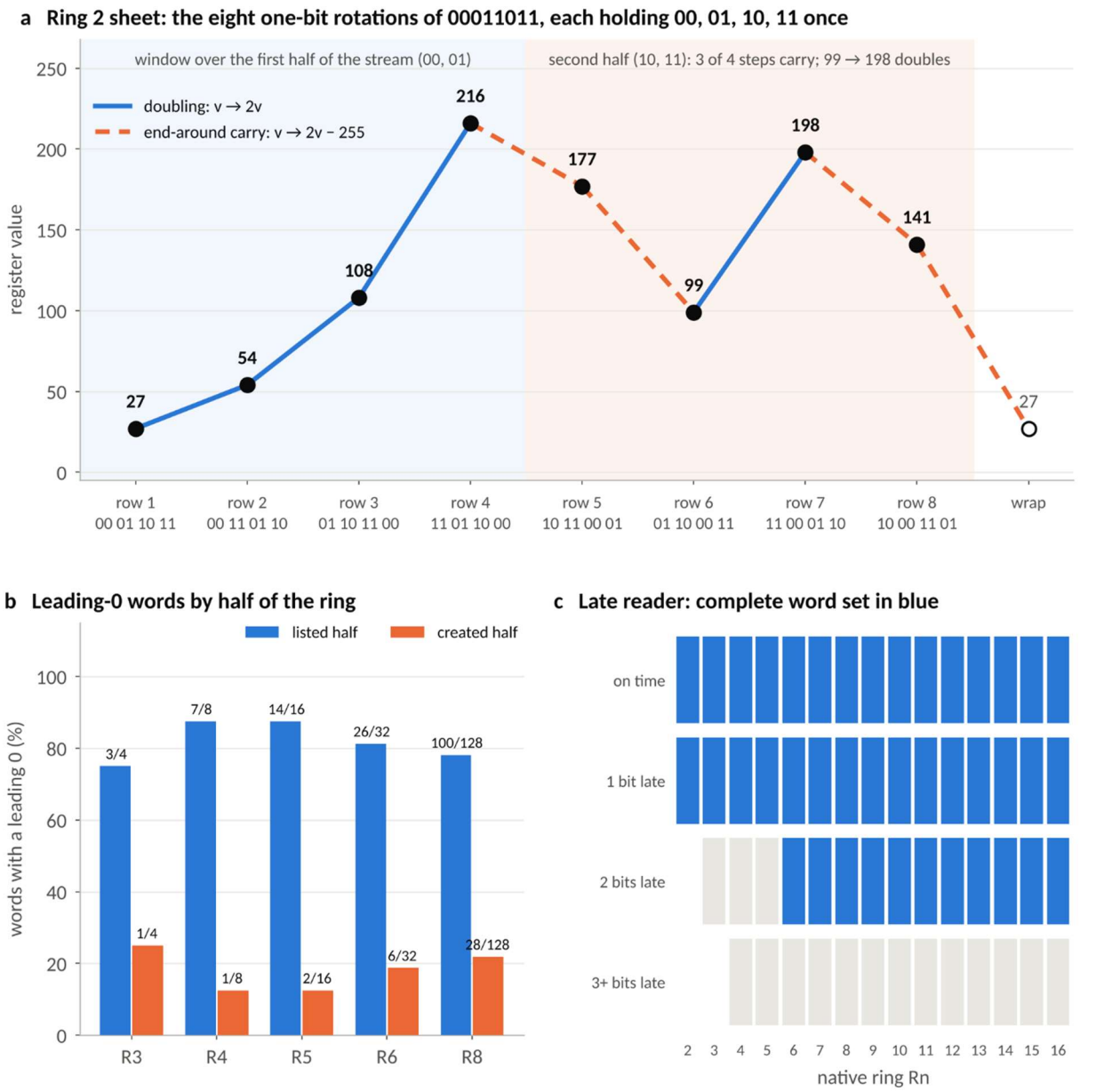


Figure 3. The Ring 2 sheet and two ring laws. (a) The eight one-bit rotations of 00011011 down the Ring 2 sheet: a row doubles (blue) while the departing bit is 0 and takes the end-around carry 2v − 255 (orange, dashed) when it is 1; in the second half three of four steps carry and 99 → 198 doubles. Every row splits into 00, 01, 10 and 11 once. (b) Words with a leading 0 in the listed and created halves of R3–R8. (c) Reading each ring stream from a later starting bit: one bit late gives a complete word set on every ring from R2 to R16, two bits late from R6 on, three or more never. Data: the nested-ring workbook (doubles.py), the ring builder, ringchk.py and ringphase.py.

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This single recurrence law accounts for every doubling flag on the Ring 2 and Ring 3 sheets of the nested-ring workbook with zero discrepancies (RUNS): on **Ring 2**, it matches **49 of 49** clockwise doubles, **56 of 56** counter-clockwise doubles, and shows that **42 of 42** non-doubles differ from $2v_t$ by *exactly* the carry; on **Ring 3**, it matches **529 of 529** clockwise doubles, **552 of 552** counter-clockwise doubles, and **414 of 414** non-doubles off by *exactly* the carry.

The carry changes the value and never the set. Every row of the Ring 2 sheet, carry rows included, splits into 00, 01, 10 and 11 exactly once, so each row holds the complete amide-nitrogen port set. The same holds for every native ring read from a later starting bit: a reader that starts one bit late reads a complete word set on every ring from R2 through R16, a reader two bits late does so from R6 on, and a reader three or more bits late never does (RUNS on R2–R16; the proof is OPEN).

Why does the first half of the ring double cleanly while the second half triggers continuous carries? Because the ring builder places the smaller numerical member of each partner pair into the **listed half** (which consequently starts predominantly with a leading 0) and places the partner images into the **created (reflected) half** (which consequently starts predominantly with a leading 1):

Native Ring Dimension	Words With Leading 0: Listed Half (Forward Code)	Words With Leading 0: Created Half (Reflected Output)
Ring R_3 ($N = 8$)	**3 of 4** (75.0%)	**1 of 4** (25.0%)
Ring R_4 ($N = 16$)	**7 of 8** (87.5%)	**1 of 8** (12.5%)
Ring R_5 ($N = 32$)	**14 of 16** (87.5%)	**2 of 16** (12.5%)
Ring R_6 ($N = 64$)	**26 of 32** (81.3%)	**6 of 32** (18.8%)
Ring R_8 ($N = 256$)	**100 of 128** (78.1%)	**28 of 128** (21.9%)

Doubling is reading the listed half: forward translation without carry. Breaking the double is reading the created half: every leading-1 step forces an end-around carry, three of the four steps there, while 99 → 198 doubles. This points to the link that would close the loop between the binary rings and macromolecular folding, and that link is OPEN: **a structural crease in a protein as an arithmetic carry event**, where a forward translation run overflows its local capacity because its leading bit has turned to 1, forcing the stream to wrap around into a reflected return run.

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Radial Address Trees vs. Cell Values and the NEXUS Transition CPU

To make recursive ring navigation executable at scale without enumerating exponentially large rings (2^n), the **NEXUS Transition CPU (v0.6)** represents processor state purely as a cursor tuple STATE = (LEN,ADDRESS) and executes native instructions (PEEK, STEP, FACE, DROP, and LIFT) directly in address space. Verified across **4,088** rank/unrank checks and tested up to **1,024-bit** addresses (R_{1024}) with zero ring reconstruction, executing DROP from LEN $= 8$,ADDR $= 128$ (WORD=EF) projects cleanly to lower-resolution LEN $= 7$,ADDR $= 0$ with sheet bit SHEET $= 1$, and LIFT restores LEN $= 8$,ADDR $= 128$ (RUNS): the lower address wraps while the sheet bit flips.

When testing how information propagates radially outward from inner rings to outer rings ($R_n \rightarrow R_{n+1}$), a critical separation emerged between **radial cell addresses** and **stored word values**. On the radial address tree, parent cell address a splits exactly into child addresses $2a$ (leading branch) and $2a + 1$ (trailing branch), with the facing operator exchanging the two branches identically at every depth (RUNS). However, checking whether the native binary *word* stored at parent cell a is preserved as a substring inside its two radial children showed rapid decay beyond R_3 (DEAD END for radial word inheritance):

Radial Ring Transition	**Parent Word Kept in Both Children**	**Parent Word Kept in At Least One Child**	**Empirical Hairpin Return-Arm Word Check (3 & 4 Residues Outward)**
$R_2 \rightarrow R_3$	**3 / 4** (75.0%)	4 / 4 (100.0%)	**3-Residue Palindrome Arm:** Return = Mirror (J) **0.48** vs. Complement (C) **0.10** (Chance = 0.12).
$R_3 \rightarrow R_4$	**3 / 8** (37.5%)	8 / 8 (100.0%)	**3-Residue Non-Palindrome Arm:** Return = Mirror (J) **0.11** vs. Complement (C) **0.09** (Chance = 0.12).
$R_4 \rightarrow R_5$	**7 / 16** (43.8%)	15 / 16 (93.8%)	**4-Residue Non-Palindrome Arm:** Return = Mirror (J) **0.29** vs. Complement (C) **0.07** (Chance = 0.06).
$R_6 \rightarrow R_7$	**24 / 64** (37.5%)	54 / 64 (84.4%)	**4-Residue Palindrome Arm:** Return = Mirror (J) **0.12** vs. Complement (C) **0.04** (Chance = 0.06).
$R_9 \rightarrow R_{10}$	**50 / 512** (9.8%)	126 / 512 (24.6%)	**Conclusion:** Real hairpins return as **Mirror (J)** routes where the words follow the

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Radial Ring Transition	Parent Word Kept in Both Children	Parent Word Kept in At Least One Child	Empirical Hairpin Return-Arm Word Check (3 & 4 Residues Outward)
			sheet face (0.48 and 0.29), not complements.

This pair of results fixes the exact nature of molecular memory. Word values stop tracking radially past R_3 because static values were never what carried inheritance; **the memory of a protein is the spatial location (address) reached by a unique cumulative history of fork bits (leading or trailing at each ring transition)**. At any finite ring depth, two distinct routes never occupy the same cell (meeting only in the continuous limit on a boundary), meaning every occupied state possesses a unique execution history. Because facing on radial addresses simply swaps leading and trailing branches at every fork without inspecting cell content, the return arm of a real protein hairpin emerges as the mirror (J) of the forward arm (0.48 on 3-residue palindromes and 0.29 on 4-residue non-palindromes against 0.12 and 0.06 chance; 3-residue non-palindromes sit at chance, 0.11 against 0.12).

Decompiling a folded 3D protein back to its 1D sequence strips away the accumulated carry history, collapsing unique spatial addresses back into an undifferentiated seed ("just another chain," the way every unfertilized egg looks identical or the way Ring 3 positions remain ambiguous until the sliding window reaches 11 bits). Consequently, no local sequence-window reader can read the final 3D address off the seed alone; the address is read with whole-chain context, as the field query does in 95 of 154 chains. On the rings the same holds in exact form: from R8 through R12 the native stream fixes a position only with a window of exactly four words.

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Synthesis: Master Ledger Audit, Object-Oriented Cell Runtime, and the Grower Specification

Complete Verification Ledger of the NEXUS Fold Compiler

Synthesizing all executed lanes across the exploration logs establishes a clear boundary between verified executable components (RUNS and STANDS), falsified value-reading shortcuts (DEAD END), and the remaining links required to close full 3D compilation (OPEN):

Compiler Stage / Hypothesis	Verification Status	Novelty & Grounding Assessment
Genetic Code as 4-Line Lexer & Mitochondrial Argument (64/64)	**RUNS**	New executable compiler framing; unifies standard and vertebrate mitochondrial codes via one detail argument; reaches 24 of the 27 NCBI translation tables, and the three it misses each open a box the budget line closes.
Binary Base Shape, 16/16 Budget Rule, 8/8 Rumer Cube Fold	**STANDS**	Grounded in Jiménez-Montaño (1996), Lagerkvist (1978), Rumer (1966), and Grosjean & Westhof (2016).
4-Bit Side-Chain Route Tokens (h, e, t) & 1-Step G/P Lookahead	**RUNS**	Classic former/breaker groups derived purely from 4 topological route bits; pre-Pro/Gly torsion compression/expansion.
R_2 Atomic Slot Alphabet: Reversal Wires Chain, Complement Seals Fold	**STANDS**	New unified binary alphabet (00, 10, 01, 11) for atoms, backbone, side chains, and protonation charge.
Graph-Level Covalent Port Conservation (10 meets 01)	**STANDS**	Resolves isolated atom-byte imbalance (6/8 fail locally) via assembled peptide graph conservation (3 boundary ports).
Lane 2 → Lane 4 Recursive Operator	**RUNS**	Covalent 1001 becomes palindrome selecting torsion via C_4 and folding into carry 20; H-bond 1100 wires via J_4.

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Compiler Stage / Hypothesis	Verification Status	Novelty & Grounding Assessment
Swap & Covalent Carry (20)		
Hydropathy as Unclosed Polar Slot Count ($r = -0.81$)	**RUNS**	Re-derives empirical Kyte–Doolittle scale as a weightless window readout of water-facing 00 and 11 slots.
Weightless Field Slot Threader (Only 8.1% Decoys Beat Native)	**RUNS**	Recognizes native fold #1 in 13/31 chains with zero fitted weights; defines water (11110000) as active runtime code.
Field Query: Open Slots plus Proline Rules (95/154 Single Answers)	**RUNS**	Returns the native backbone as the single record from about 2,540 real backbone rows per E. coli sequence; proline rules supply about 5 bits and open slots about 4; recognition, not building.
Fold as Chaitin Register Allocation & Water as Spill Memory	**RUNS**	Register-allocation framing of co-translational folding: helices hold 6–8 live closures at any length, sheets climb to 34; the peak is the cutwidth of the closure graph; comparators are contact order (Plaxco et al., 1998) and N-to-C building (Ellis et al., 2010).
96% of Backbone Closures Computed by Native Ring Operators	**RUNS**	Structural census of 1,553 closures: 72.2% short successor offsets, 21.1% local reflections, 2.4% long shifts, 4.3% pointers; 95.7% run on ring operators.
Reflection Is Created: 75% of Long-Range Contacts on Reflection Runs	**RUNS**	Explains why sequence codon pairs show no seam signal (79 E. coli genes); on 146 E. coli chains the share is 0.70 and follows the sheet: antiparallel-dominant 0.86, parallel-dominant 0.31 (r = −0.96).
Sealed Sequence-Only Creator (SHA-	**RUNS**	Cleared all 3 sealed bars on structures unseen in design: L1 0.220 (bar 0.20), L2 0.082 (bar 0.06), L3 strand-crease

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Compiler Stage / Hypothesis	Verification Status	Novelty & Grounding Assessment
256: 83c7cb05…) on 79 Proteins		0.317 (bar 0.30); landing within one cell; homologs not filtered.
Parity Sectors, 16/16 R_2 Turn Phase Selector, and Beta-Bulge Slip	**STANDS / RUNS**	Complete 4-operation R_2 phase vocabulary ($m = p_{\text{old}} \oplus p_{\text{new}}$); 19/40 antiparallel seams cross parity sectors via bulges.
Bulge as the Repeat Symbol of the Face Stream	**RUNS / OPEN**	Neighbors on the same side of the sheet 0.27 at slips vs 0.08 where the register is kept; glycine 0.097 vs 0.055; the sequence trigger is OPEN.
Helix Packing Face Words Are $E_2(x) = x \parallel P_2(x)$ Words on R_4	**RUNS**	Touching helix residues occur in adjacent pairs (rotations of 1100), satisfying equivariant lift $P_4(E_2(x)) = E_2(P_2(x))$.
Sequence-Side R_4 Face Selectors v0.1 (d8368d06…) & v0.2 (e3dc59d9…)	**DEAD END**	Falsified (0/15 and 2/31 creases bound); raw sequence carbon bits rarely form exact 1100 without phase precession.
Spreadsheet Doubling Break as Ones' Complement End-Around Carry	**RUNS**	Exact law on every doubling flag of the Ring 2 and Ring 3 sheets; the listed/created split carries the doubling/carry split; crease as carry in proteins is OPEN.
Late Reader: Complete Word Set One Bit Late	**RUNS**	Every native ring R2–R16 read one bit late gives every word once, two bits late from R6 on; every Ring 2 sheet row is a complete port set.
Radial Address Tree Inheritance vs. Radial Word Content Decay	**RUNS / DEAD END**	Radial addresses split/invert exactly (RUNS), while word content decays to 9.8% by R_{10} (DEAD END); memory is address route.

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Systems Conclusion: The Co-Translational Grower and the Object-Oriented Cell Runtime

Zooming out to the cellular runtime explains *why* biological matter compiles sequences into three-dimensional objects at all. A living cell is a dense chemical soup where nearly every reaction is thermodynamically permitted at slow background rates—an unscheduled equilibrium where everything slowly drifts toward everything else. Because a cell cannot build physical walls between every reacting molecule, the only way to select a specific causal pathway without forbidding others is through **relative transformation speed**. By accelerating one specific chemical transformation by factors from 10^6 up to 10^{17} (orotidine 5′-monophosphate decarboxylase; Radzicka & Wolfenden, 1995) over background noise, an enzyme acts as a **thread priority scheduler**: it forces the selected reaction to finish and hand its product to the next pathway step before background equilibrium can scatter the intermediates, generating causal ordering (time) and signal-to-noise separation purely through rate control.

In this object-oriented execution model, the gene functions as the static class definition while the ribosome operates as the object factory—an instance factory itself constructed from compiled RNA and protein instances. The folded protein is the compiled binary object instance, and its three-dimensional binding site—the specific geometric *hole* or negative space left behind by the folding build—is the object's **duck-typed public method signature**, accepting any substrate whose molecular shape fits the cavity without calling the protein by name. Depending on their architecture, enzymes execute as stateless pure-function services, allosteric receptors act as stateful objects with event handlers, transcription factors operate as feature-flag dispatchers, and structural proteins serve as static data structs. Crucially, the caller that invokes protein compilation is the **metabolic gap** itself—exemplified by the *E. coli lac* operon (Jacob & Monod, 1961), which runs on two conditions. Allolactose, made from lactose by the few β-galactosidase molecules present before induction, releases the LacI repressor brake, and the absence of glucose, read through cAMP and CAP, opens the promoter. Lactose present and glucose absent: the gap, read through the field's two conditions, calls the ribosomal factory to instantiate the missing digestive enzyme.

To advance the NEXUS fold compiler from recognizing folds and projecting 2D contact seams to growing complete 3D coordinates de novo, the next build must replace static sequence-window parsers with the **Co-Translational Address Grower**. Because the linear sequence is only the first operand (2 + =), the grower must emit residues sequentially in ribosomal loader order into the 11110000 water field, allow Glycine and Proline routing devices to hold local segments open against immediate $i + 4$ helical closure, and execute each field-forced closure as an **end-around carry event** ($v_{t+1} = 2v_t - (2^W - 1)b_t$) that appends a fork bit to the residue's stacked radial address. Closing the six remaining OPEN links—detecting the sequence crease directly as a leading-bit carry transition, adding the sub-atomic **nucleus lane** beneath R_2 to separate charged $-NH_3^+$ (1111) from neutral methyl $-CH_3$ (1111) and weak sulfur lone pairs in Cysteine/Methionine from oxygen, aligning the 3.6-residue sequence phase reader to the R_4 1100 helix face word, resolving the 28 non-adjacent partner runs via 3D burial and G/P hold duration, triggering the one-sided beta-bulge parity slip (B_i, B_j) from partner homelessness across the seam, and linking Serine's complementary codon islands (TCN and AGY) to N-cap gate switching—will complete the deterministic compilation pipeline from raw genomic bits to three-dimensional macromolecular coordinates.

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Data and Code

Every run in this paper is a script over public data: Protein Data Bank coordinates for the 30 training structures, and 155 *E. coli* chains matched to their genes through UniProt and the European Nucleotide Archive, with every translated coding sequence equal to its UniProt protein. The NCBI translation tables are used as distributed with Biopython (Cock et al., 2009). The sealed predictions travel with their files: the creator seal (SHA-256 83c7cb05…) and the v0.1 and v0.2 face-selector preregistrations (d8368d06…, e3dc59d9…), together with the scripts, the chain lists and the nested-ring workbook, accompany this paper, so that every hash and every count in the ledger can be run again.

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github.com/QuHarmonics/The-Nexus-Harmonic-Reality info@quharmonics.com